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 FYP1 - Final Report

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Submission ID

trn:oid:::3618:127523760

Submission Date

Feb 4, 2026, 6:58 PM GMT+8

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File Name

FYP literature review unlinked (after word) turnitin.pdf

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**FACULTY OF
COMPUTING
& INFORMATICS**

FYP01-DS-T2530-0860
Autonomous Driving Agent Training Using Deep
Reinforcement Learning in CARLA Simulator

242UC243LS

NG JUN XUAN

Chapter 1: Introduction

1.1. Background of the Research

Autonomous driving has become a major trend in the automotive industry, with most leading car manufacturers integrating autonomous capabilities into their vehicles (Huynh & Nguyen, 2024, p. 1). However, despite rapid progress, autonomous driving systems still face several challenges. Experiments on the CARLA simulation platform show that rule-based control methods are vulnerable in dynamic and complex environments, especially when the sensing module is limited or environmental conditions change, their performance degrades significantly (Dosovitskiy et al., 2017, p. 7). In addition, the training of deep learning-based driving models requires large amounts of labeled data, which are often difficult, expensive, or unsafe to collect in real-world scenarios.

1.2. Problem Statement

Autonomous driving technology still faces significant driving safety challenges in complex, high-risk driving scenarios such as intersections, ramp merging, and unsignalized intersections (Luo et al., 2025, p. 1). While reinforcement learning (RL) has shown potential in autonomous driving decision-making and planning tasks, the safety performance differences of different RL algorithms (such as PPO (Proximal Policy Optimization), SAC (Soft Actor-Critic), and DDPG (Deep Deterministic Policy Gradient)/TD3 (Twin Delayed Deep Deterministic Policy Gradient)) when using the Transformer architecture remain unclear (Yun & Cao, 2024, p. 6). Currently, there is a wealth of research on autonomous driving, but the models, algorithms, and methods used vary, making it difficult to directly determine which combination is best suited for high-risk scenarios. Therefore, researchers and system designers face difficulties in selecting appropriate RL algorithms to improve the safety of autonomous driving systems.

1.3. Research Purpose

The purpose of this review is to comprehensively analyze and synthesize research on the safety of Transformer-based reinforcement learning algorithms in high-risk autonomous driving scenarios. Specifically, this review purpose is to:

- Classify commonly used Transformer architectures in autonomous driving tasks.
- Compare the safety performance of these algorithms (PPO, SAC, DDPG/TD3).
- Rank the risks of various scenarios based on literature data.

This review can serve as a foundational analysis, providing a reference for the design of FYP2.

1.4. Objectives of the Research

1. To systematically identify and categorize the most commonly used Transformer-based model and reinforcement learning algorithms in autonomous driving tasks.
2. To quantitatively compare the safety performance of reinforcement algorithms under comparable driving scenarios.
3. To rank the risk levels of different high-risk autonomous driving scenarios using a consistent set of safety metrics.

1.5. Research Questions

The main research question is under a Transformer-based policy network architecture, what are the differences in safety performance among different reinforcement learning algorithms in high-risk autonomous driving scenarios?

To answer this question, this study explores the following sub-questions:

- What are some commonly used Transformer-based models in high-risk autonomous driving tasks?

- What are the differences in safety performance among different reinforcement learning algorithms in similar high-risk driving scenarios?
- What are the differences between different high-risk scenarios?

1.6. Project Scope

This study aims to systematically evaluate the safety performance of different reinforcement learning algorithms under the Transformer architecture in high-risk autonomous driving scenarios.

1.6.1. Technical Scope

This study focuses on model architecture, algorithm performance, scenario risk, and safety metrics.

- This study uses Transformer-based models as the core component of the state encoder or policy network.
- This study primarily compares the safety performance differences of PPO, SAC, and DDPG/TD3 under the same Transformer architecture.
- This study will focus on the most common high-risk scenarios identified in the collected literature. These include unsignalized intersections, merging ramps, roundabouts, and highways.
- This study mainly uses quantitative safety metrics such as collision rate and success rate for evaluation.

1.6.2. Data Sources and Screening Scope

- This literature review only includes the top 100 papers in Google Scholar and all papers published in the IEEE Xplore database between 2019 and 2025.
- Only papers containing the Transformer architecture, specific reinforcement learning algorithms (PPO, SCA, DDPG/TD3), and

safety-related metrics are considered. Papers involving non-autonomous driving scenarios, such as drones and driver state detection, will not be considered. Review articles and articles that are unavailable are not within the scope of this literature review.

1.6.3. Output and Contribution Scope

- Compare the safety performance of various algorithms under the same transformer model
- Rank the driving scenarios of each paper by risk level.
- Summarize the types of transformers used in autonomous driving.

1.7. Significance of the Research

This study, through a systematic literature review and quantitative comparative analysis, explores the safety performance of different reinforcement learning algorithms under the Transformer architecture in high-risk autonomous driving scenarios. It has the following important theoretical and practical implications:

1.7.1. Theoretical Significance

- Systematic Study of Algorithm Comparison: This study provides a comparative framework through a unified architecture and evaluation criteria, facilitating a more objective assessment of algorithm performance differences.
- Constructing a Quantitative System for Scenario Risk Assessment: Based on data from multiple empirical studies, this study constructs a risk level ranking model for high-risk autonomous driving scenarios (such as intersections, lane changes, roundabouts, and highways), providing a theoretical

basis for scenario complexity classification and assessment in subsequent research.

- Promoting the Development of Transformer-RL Fusion Theory: By systematically reviewing different variants and applications of the Transformer in autonomous driving (such as ViT, Decision Transformer, and Multimodal Transformer), this study provides architectural choices for future research.

1.7.2. Practical Significance

- Providing a Basis for Autonomous Driving System Design and Algorithm Selection:
 - This study indicates that the safety performance of algorithms such as SAC, PPO, and DDPG/TD3 varies significantly across different high-risk scenarios. The research results provide empirical support for autonomous driving system developers in algorithm selection and optimization, contributing to improved system safety and reliability.
- Guiding Simulation Test Design in High-Risk Scenarios:
 - Based on the scenario risk assessment results, this study provides priority suggestions for test scenario design in simulation platforms such as CARLA, helping testing teams focus on high-risk scenarios and improve testing efficiency and coverage.

Chapter 2: Literature review

2.1. Introduction

Reinforcement learning has demonstrated superior ability to handle complex scenarios in autonomous driving compared to rule-based systems. Traditional rule-based methods often struggle to cope with the complexities of real-world traffic because they rely on manually designed driving decisions by humans (G. Zhao et al., 2025, p. 1).

This chapter details the literature review process and provides a comprehensive overview of the latest research progress combining the Transformer architecture with various reinforcement learning algorithms, such as PPO, SAC, and DDPG/TD3. It also provides formal definitions for complex driving scenarios (including lane changes, roundabouts, and intersections) that can serve as testbeds for these agents.

2.2. Systematic Literature Review Methodology

2.2.1. Search Strategies

- Database: Google Scholar, IEEE Xplore
- Years: 2019-2025
- Keyword: ("Transformer") AND ("Reinforcement Learning" OR "Deep Reinforcement Learning" OR DRL OR RL OR "Q-learning" OR "Policy Gradient") AND ("Autonomous Driving" OR "Autonomous Vehicle" OR "Self-driving" OR "Intelligent Vehicle" OR CAV) AND ("Collision Avoidance" OR "Time to Collision" OR TTC OR Safety OR "Near-miss" OR Accident OR Crash OR "Risk Assessment")

2.2.2. Include criteria

- Research must use a Transformer or a Transformer-based model as its core model.

- Papers must cover at least one specified reinforcement learning algorithm, including PPO, SAC, or DDPG/TD3.
- Quantifiable safety assessment data must be provided.

2.2.3. **Exclude criteria**

- All review articles, surveys, tutorials, or systematic reviews are excluded because they do not provide original experimental data.
- Research on non-autonomous driving, such as driver fatigue detection, is excluded. Research applied to off-road traffic scenarios (e.g., off-road vehicles) is also excluded.

2.2.4. Screening process

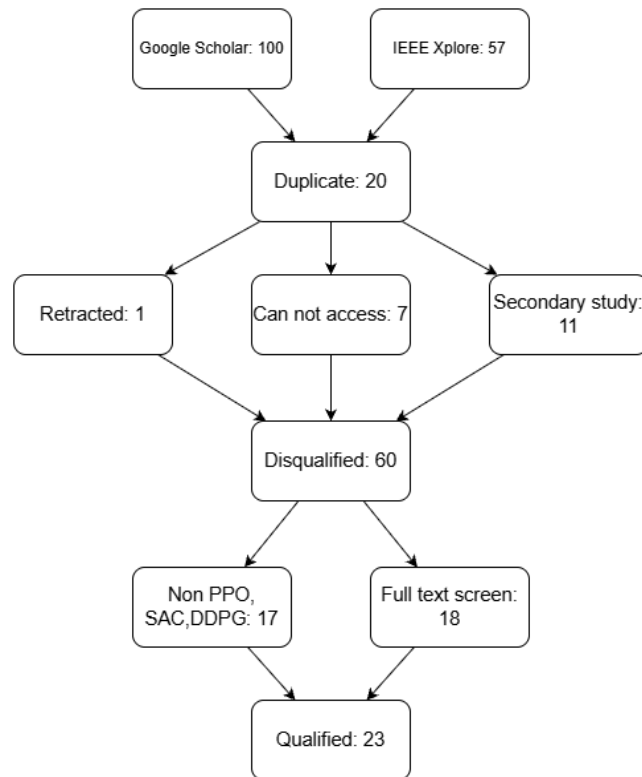


Figure 2.1 Screening Flow chart

2.3. Specific Applications of Transformer-RL in Autonomous Driving

In recent years, the Transformer architecture, with its self-attention mechanism, has revolutionized the field by demonstrating exceptional capabilities in handling long-range dependencies and complex relationships in modeled data (Doroudian & Taghavifar, 2024, p. 1).

2.3.1. State Encoder Design (Multimodal Transformer)

In driving tasks, simple models like convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are unsuitable because autonomous driving relies on data from various sensors, such as camera images and LiDAR point clouds. These simple models cannot handle this diverse range of data formats and sizes.

The Transformer architecture overcomes these limitations through its self-attention mechanism (Vaswani et al., 2023). Transformers can compute relationships between different input data without considering the different data formats, and this mechanism can handle data with large time spans well. In multimodal state encoding, this capability enables cross-modal attention, allowing features from one modality (e.g., visual objects) to directly focus on relevant features in another modality (e.g., vehicle states such as steering, linear velocity). For example, the DAD-RL framework proposed by (Chowdhury et al., 2024) adopts a multimodal input design, including historical vehicle trajectories and bird's-eye view (BEV) contextual information. Specifically:

- Historical trajectory modality: contains the state sequence (position, heading, speed, lane) of the vehicle and surrounding vehicles in the past 2.1 seconds, which is spatiotemporally encoded using LSTM and attention mechanisms;
- BEV image modality: contains a drivable area map and a waypoint map, which extracts contextual features using CNN.

Driving decisions inherently depend on both current and historical states. Although traditional sequence processing models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks can capture temporal dependencies, they are limited in handling long-range sequences and suffer from slow training due to their sequential nature. In contrast, Transformers leverage parallelizable self-attention mechanisms, enabling efficient processing of long temporal spans (Fan et al., 2024).

2.3.2. Policy Network Architecture (Transformer as Policy/Actor)

Compared to MLPs, Transformers are better suited as policy networks. MLPs lack explicit mechanisms for modeling temporal dependencies. Transformers, however, can analyze

relationships between data spanning long time periods through self-attention mechanisms.

In the paper (Doroudian & Taghavifar, 2024), the authors incorporated Transformers as part of a policy network and combined it with the PPO algorithm. The PPO Transformer (TPPO) achieved a 98% success rate and only a 2% collision rate, while the non-Transformer PPO achieved a 42% success rate and a 12% collision rate. The safety of PPO with Transformers is significantly improved compared to the non-Transformer PPO.

2.4. Comparative Applications of RL Algorithms in Autonomous Driving Decision-Making

2.4.1. Application Research of PPO

In traditional reinforcement learning frameworks, policy gradient methods exhibit extremely high sensitivity to the learning rate. On the one hand, an excessively small learning rate leads to slow parameter updates, significantly increasing the demand for training samples; on the other hand, an excessively large learning rate can cause disastrous consequences. Since an agent's actions directly alter its subsequent states in reinforcement learning, a large policy update can cause the agent to enter previously unexplored disadvantageous regions, resulting in a sharp decline in the quality of collected data and ultimately inducing policy collapse.

To overcome this problem, (Schulman, Levine, et al., 2017) proposed Trust Region Policy Optimization (TRPO). The core of TRPO lies in using Kullback-Leibler divergence to rigorously measure and limit the difference between new and old policies, ensuring that each update is within a safe "trust region." However, the theoretical elegance of TRPO comes at the cost of extremely high computational cost; the calculation of

its second derivative (Hessian matrix) is extremely complex, and its implementation is difficult.

Inspired by TRPO, (Schulman, Wolski, et al., 2017) proposed Proximal Policy Optimization (PPO). The core innovation of PPO lies in the introduction of a "Clipped Surrogate Objective," which restricts the probability ratio of the new and old policies to a predefined interval (typically $[1 - \epsilon, 1 + \epsilon]$), thus achieving stability similar to TRPO in a simpler first-order derivative manner. This mechanism not only significantly lowers the implementation threshold of the algorithm but also allows for multiple training iterations on the same batch of data, greatly improving sample efficiency.

Experiments demonstrate that PPO exhibits superior performance in both continuous control and discrete action space tasks (Schulman, Wolski, et al., 2017). Compared to methods such as A2C, PPO maintains extremely high implementation flexibility while ensuring convergence speed and final performance. With its high stability and wide applicability, PPO has become one of the most highly regarded core algorithms in the fields of autonomous driving, robot control, and game AI.

2.4.2. Application Research of SAC

Soft Actor-Critic (SAC) proposes a stochastic actor-critic algorithm using a maximum entropy reinforcement learning framework and an off-policy learning mechanism, which improves sample efficiency while maintaining training stability (Haarnoja et al., 2018). SAC comprises two main neural network components (Haarnoja et al., 2018):

- Actor network: Outputs the distribution of actions (e.g., Gaussian distribution) with an explicit entropy term to encourage exploration (Haarnoja et al., 2018).

- Two Q-function networks (Critics): Utilized to reduce the positive bias in value function estimation through a clipped double-Q trick (Haarnoja et al., 2018).

2.4.3. Application Research of DDPG

Reinforcement learning methods for discrete action spaces, such as DQN, cannot be directly applied to continuous action spaces because the "maximization" operation of continuous actions requires complex optimization at each step, resulting in high computational cost and instability. DDPG combines the ideas of Deterministic Policy Gradient (DPG) and Deep Q-Networks (DQN) to propose an off-policy, model-agnostic actor-critic algorithm suitable for continuous action spaces (Lillicrap et al., 2019).

DDPG is based on the DPG framework, where:

- Actor: Outputs deterministic continuous actions.
- Critic: Evaluates the Q-values of state-action pairs.

By using a replay buffer and target networks, DDPG borrows the stable training mechanism of DQN, achieving efficient learning in continuous action spaces (Lillicrap et al., 2019).

2.4.4. Application Research of TD3

A problem exists in the DQN algorithm: Q-value overestimation. Since DDPG inherits from DQN, it also inherits this problem. This issue arises because the DQN critic overestimates the Q-value when evaluating actions due to noise, and actors tend to learn those overestimated actions, ultimately leading the policy to suboptimal solutions or even training failure.

To address this problem, TD3 was proposed. This algorithm introduces three main features:

- Delayed actor updates: This allows the critic to stabilize before the actor learns actions closer to the actual Q-value.
- Adding two independent critics: This ensures that the smallest Q-value is selected. While this might still lead the actor to learn actually underestimated actions, (Fujimoto et al., 2018) show that learning underestimated actions is safer than learning overestimated ones.
- Target policy smoothing: Adding a small amount of noise to the target action smooths the Q-value estimation and improves robustness.

2.5. Definition and Research of High-Risk Autonomous Driving Scenarios

2.5.1. Classification and Definition of High-Risk Scenarios

To cultivate robust decision-making in autonomous agents, exposure to high-risk scenarios during training is imperative. Analogous to the adage that calm seas do not produce skilled sailors, a benign training environment fails to equip agents with the necessary skills to handle unexpected real-world situations.

- Merging
 - Definition: A scenario where vehicles, without traffic signal guidance, need to observe, predict, and negotiate right-of-way with traffic flows from multiple directions.
 - Research Focus: Conflict resolution, intent prediction, and game theory decision-making.
- Roundabout

- Definition: A multi-lane roundabout where vehicles need to merge, cycle, and exit within a continuous traffic flow.
 - Research Focus: Cycle gap determination, lane keeping, and exit decision-making.
- Turn intersection
 - Definition: A turn intersection includes: unsignaled intersections, four-way intersections, left turns, and unprotected left turns. I categorize them all as turn intersections. It is a scenario where vehicles, without traffic signals, must observe, predict, and negotiate right-of-way with multi-directional traffic flows (including oncoming vehicles, pedestrians, and non-motorized vehicles).
 - Research Focus: Conflict resolution, intent prediction, and game theory decision-making.
- Highway
 - Definition: A high-speed, multi-lane environment. In some papers, the highway scenario may include overtake tasks.
 - Research Focus: Course deviation, maintaining distance, performance in severe weather.

2.5.2. Risk Assessment Index System

Each paper uses different metrics to measure the performance of its model and algorithm. Below is a summary of the most commonly used metrics across these papers. While some papers may use the same metrics, their measurement methods may vary slightly.

- Collision Rate: This is the most frequently used and most direct evaluation metric, calculating the proportion of test scenarios where collisions occur.

- **Collisions Time:** This refers to the total number of collisions that occur when the autonomous vehicle completes a navigation task in each test scenario.
- **Success Rate:** This is the second most frequently used metric. This metric may include not only the task completion rate without collisions but also other factors such as reaching the destination within a specified time.
- **Lane Departure Rate:** Commonly used to evaluate whether a vehicle can stably maintain its position near the center of the lane, avoiding excessive swaying or deviation. This metric is usually measured in centimeters (cm).

Chapter 3: Algorithm Performance Comparison

3.1. Introduction

This chapter aims to compare the safety performance of different deep reinforcement learning (DRL) algorithms in autonomous driving decision-making tasks through a systematic analysis of experimental data from existing literature. Although algorithms such as PPO, SAC, and DDPG/TD3 each have their own theoretical focus, their actual safety and effectiveness under the complex Transformer architecture still need to be verified through quantitative metrics. This chapter will focus on the horizontal and vertical performance evaluation of various mainstream algorithms based on key indicators such as collision rate, collision time, and success rate.

3.2. Comparative Analysis of PPO and SAC

Table 3.1 Performance Comparison between PPO and SAC

Paper title	PPO Performance	SAC Performance	Conclusion (Low performance < High performance)
Augmenting Reinforcement Learning With Transformer-Based Scene Representation Learning for Decision-Making of Autonomous Driving (Liu et al., 2024)	Collision rate: Min: 34% Mean: 51.2% Max: 64%	Collision rate: Min: 22% Mean: 35% Max: 52%	PPO < SAC
DAD-RL for Autonomous Decision-Making in Dynamic Environment (Chowdhury et al., 2024)	Collision rate: Min: 34% Mean: 51.2% Max: 64%	Collision rate: Min: 22% Mean: 35% Max: 54%	PPO < SAC

Table 3.1 (continued)

Efficient Maximum Entropy Reinforcement Learning Framework Guided by Expert Policies: An Application in Autonomous Driving (L. Gao et al., 2025)	Success rate:	Success rate:	PPO < SAC
	Min: 36%	Min: 65%	
	Mean: 37%	Mean: 68.5%	
	Max: 38%	Max: 72%	
Improving Reinforcement Learning with Expert Demonstrations and Vision Transformers for Autonomous Vehicle Control (Elallid et al., 2024)	Collision rate:	Collision rate:	PPO < SAC
	20%	6%	
	Success rate:	Success rate:	
	80%	94%	
SoftCTRL: Soft conservative KL-control of Transformer Reinforcement Learning for Autonomous Driving (Huynh & Nguyen, 2024)	Collision time:	Collision time:	PPO < SAC
	Mean: 20.87 $\pm$ 1.14	Mean: 1.87 $\pm$ 0.25	

Table 3.1 shows all the performance data that the papers include both PPO and SAC algorithms. In the data from these five papers, the SAC algorithm is significantly more secure than the PPO algorithm.

(Liu et al., 2024), PPO has a minimum collision rate of 34%, an average of 51.2%, and a maximum collision rate of 64%, while SAC has a minimum collision rate of 22%, an average of 35%, and a maximum collision rate of 52%. This shows that SAC's collision rate is lower than PPO's collision rates. It is noteworthy in this table that the data from (Liu et al., 2024) and (Chowdhury et al., 2024) are very similar. This is because (Chowdhury et al., 2024) explicitly built upon (Liu et al., 2024)'s framework, using it as a baseline for comparison. This

represents standard academic practice where new methods are evaluated against established benchmarks.

Regarding success rate, (L. Gao et al., 2025) shows that PPO's success rate is also lower than SAC's in terms of minimum, average, and maximum. (Elallid et al., 2024) used two metrics: in terms of collision rate, PPO (20%) was significantly higher than SAC (4%), while in terms of success rate, PPO (80%) and SAC (94%), both metrics showed that PPO was inferior to SAC. (Huynh & Nguyen, 2024) used a rarer metric: in terms of the number of collisions, PPO was nearly 11 times that of SAC.

3.3. Comparative Analysis of SAC and DDPG/TD3

Table 3.2 Performance Comparison between SAC and DDPG/TD3

Paper title	SAC Performance	DDPG/TD3 Performance	Conclusion (Low performance < High performance)
Improving policy training for autonomous driving through randomized ensembled double Q-learning with Transformer encoder feature evaluation (Fan et al., 2024)	Success rate: Min: 16.2% Max: 16.4%	Success rate: Min: 16.3% Max: 16.4%	SAC $\approx$ DDPG/TD3

Table 3.2 shows performance data for both the SAC and TD3 (DDPG) algorithms. In (Fan et al., 2024), the performance of SAC and TD3 is almost identical.

3.4. Comparative Analysis of Merging and Intersection Scenarios DDPG/TD3 and PPO

Table 3.3 Performance Comparison between DDPG/TD3 and PPO

Paper title	DDPG/TD3 Performance	PPO Performance	Conclusion (Low performance < High performance)
Autonomous Intersection Management via Prior-Enhanced Multi-Agent Constrained Decision Transformer (R. Zhao et al., 2025)	Collision rate: Min: 44.37% Mean: 58.98% Max: 69.86%	Collision rate: Min: 60.25% Mean: 66.95% Max: 77.99%	PPO < DDPG/TD3
Constraint-Guided Behavior Transformer for Centralized Coordination of Connected and Automated Vehicles at Intersections (Zhao et al., 2024)	Collision rate: Min: 66.73% Mean: 68.295% Max: 69.86%	Collision rate: Min: 64.265 Mean: 71.125% Max: 77.99%	PPO < DDPG/TD3

In (R. Zhao et al., 2025; Zhao et al., 2024), the average collision rates of DDPG/TD3 were 58.98% and 68.295%, respectively, while those of PPO were 66.95% and 71.125%. These results indicate that PPO performs worse than DDPG/TD3.

3.5. Comparative Analysis of Merging and Intersection Scenarios PPO, SAC and DDPG/TD3

Table 3.4 Performance Comparison between PPO, SAC and DDPG/TD3

Paper title	PPO Performance	SAC Performance	DDPG/TD3 Performance	Conclusion (Low performance < High performance)
Safety-constrained Reinforcement Learning with Interaction-aware for Decision-making of Autonomous Driving (Zhang et al., 2025)	Success rate: Min: 42% Mean: 51.33% Max: 58%	Success rate: Min: 33% Mean: 38.7% Max: 48%	Success rate: Min: 7% Mean: 18.7% Max: 37%	DDPG/TD3 < SAC SAC < PPO DDPG/TD3 < PPO

(Zhang et al., 2025) is the most direct in determining which algorithm is optimal because it directly compares the three. (Zhang et al., 2025) shows that PPO is the best, with an average success rate of 51.33%, followed by SAC with an average success rate of 38.7%, and finally DDPG/TD3 with an average success rate of 18.7%.

3.6. Analysis summary

Table 3.5 Comparative Performance Analysis of Reinforcement Learning Algorithms

No	Conclusion (Low performance < High performance)	Support paper
1	$PPO < SAC$	(Liu et al., 2024) (Chowdhury et al., 2024) (L. Gao et al., 2025) (Elallid et al., 2024) (Huynh & Nguyen, 2024)
2	$SAC \approx DDPG/TD3$	(Fan et al., 2024)

Table 3.5 (continued)

3	$PPO < DDPG/TD3$	(R. Zhao et al., 2025) (Zhao et al., 2024)
4	$DDPG/TD3 < SAC$	(Zhang et al., 2025)
5	$SAC < PPO$	(Zhang et al., 2025)
6	$DDPG/TD3 < PPO$	(Zhang et al., 2025)

Table 3.5 summarizes the performance comparisons of the PPO, SAC, and DDPG/TD3 algorithms in the main studies. This table reveals two contradictions. The first is the comparison between PPO and SAC. While most studies (5 supporting papers) agree that $PPO < SAC$, indicating that SAC has higher efficiency in autonomous driving tasks, a few studies (1 paper) reach the opposite conclusion: $SAC < PPO$. The second is the comparison between PPO and DDPG/TD3. Two papers support the claim that $PPO < DDPG/TD3$, while only one paper supports $PPO > DDPG/TD3$.

In conclusion, there are a total of 8 papers covering the PPO algorithm, 7 papers covering the SAC algorithm, and 3 papers covering the DDPG/TD3 algorithm.

This study, through a systematic literature review, finds that PPO, SAC, and DDPG/TD3 do not have an absolute superiority or inferiority in autonomous driving decision-making tasks. Their performance is highly dependent on task attributes, state representation complexity, reward function design, and environmental stochasticity. For example, PPO performs better in safety-constrained tasks requiring stable policy updates, while SAC is more advantageous in scene representation learning where exploration efficiency is crucial. Therefore, algorithm selection should be based on a thorough analysis of task characteristics.

Chapter 4: High-Risk Scenario Risk Assessment

4.1. Introduction

(Lu et al., 2025, p. 1) points out that autonomous driving tasks have several characteristics, including high vehicle density, diverse traffic participants, and complex traffic rules. These factors will affect the safety and robustness of autonomous driving algorithms. This chapter summarizes and extracts data from all the papers and uses this data to rank the risk levels of various scenarios. This chapter will use logical deduction to construct a comprehensive ranking of scenario risk levels.

4.2. Comparative Analysis of Merging and Intersection Scenarios

Table 4.1 Risk Assessment between Merging and Intersection

Paper title	Merging risk assessment	Intersection risk (Left turn) assessment	Conclusion (Low risk < High risk)
Efficient Maximum Entropy Reinforcement Learning Framework Guided by Expert Policies: An Application in Autonomous Driving (L. Gao et al., 2025)	Success rate:	Success rate:	Intersection (Left turn) < Merging
	Min: 38%	Min: 36%	
	Mean: 69.14%	Mean: 72%	
	Max: 92%	Max: 96%	

Table 4.1 compares the performance of autonomous driving in two high-risk scenarios—merging and intersections. As you can see, the average success rate at intersections is 72%, and at merging it's 69.14%, with little difference between the two. This indicates that in this study, the risks of these two scenarios are comparable, but merging is slightly more dangerous.

4.3. Comparative Analysis of Roundabout and Intersection

Table 4.2 Risk Assessment between Roundabout and Intersection

Paper title	Roundabout risk	Intersection (Right	Conclusion
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	assessment	turn) risk assessment	(Low risk < High risk)
Improving Reinforcement Learning with Expert Demonstrations and Vision Transformers for Autonomous Vehicle Control (Elallid et al., 2024)	Collision rate: 6%	Collision rate: 32%	Roundabout < Intersection (Right turn)
	Roundabout risk assessment	Intersection risk assessment	Conclusion (Low risk < High risk)
Autonomous Driving Agents for Safe Road Junctions Through Deep Reinforcement Learning (Yun & Cao, 2024)	Collision rate: Min: 7.2% Mean: 14.12% Max: 20.47%	Collision rate: Min: 56.88% Mean: 62.28% Max: 65.51%	Roundabout < Intersection

Table 4.2 compares the performance of autonomous driving models in roundabout and intersection scenarios. (Elallid et al., 2024) shows that autonomous driving models are more prone to accidents in intersection scenarios, with a collision rate of 32% compared to 6% in roundabout scenarios. Beside, (Yun & Cao, 2024) also indicates that the intersection scenario is higher risk than roundabout.

4.4. Comparative Analysis of Intersection and Highway

Table 4.3 Risk Assessment between Intersection and Highway

Paper title	Intersection risk assessment	Highway risk assessment	Conclusion (Low risk < High risk)
Empirical Evaluation of Deep Learning based Autonomous Driving System with Real-Time Scene Classification Procedure (Prasath A & Navaz K, 2025)	Lane deviation error: Min: 6.47cm Mean: 9.77cm Max: 12.4cm	Lane deviation error: Min: 6.22cm Mean: 8.96cm Max: 13.45cm	Highway < Intersection
	Intersection risk (Left turn) assessment	Highway risk assessment	Conclusion (Low risk < High risk)
Pre-trained Transformer-Enabled Strategies with Human-Guided Fine-Tuning for End-to-end Navigation of Autonomous Vehicles (Hu et al., 2024)	Collision rate: Min: 0% Mean: 21.93% Max: 58.7%	Collision rate: Min: 0% Mean: 8.84% Max: 38.7%	Highway < Intersection (Left turn)

Table 4.3 contains two papers, each using a different metric: lane deviation error and collision rate. Both papers show that the risk is higher in intersection scenarios than in highway scenarios. In (Prasath A & Navaz K, 2025), the average lane deviation error for intersections is 9.77 cm, while the average lane deviation error for highways is 8.96 cm. (Hu et al., 2024) shows an average collision rate of 21.93% for intersections, compared to 8.84% for highway scenarios.

4.5. Comparative Analysis of Merging, Roundabout and Intersection

Table 4.4 Risk Assessment between Roundabout and Intersection

Paper title	Merging risk assessment	Roundabout risk assessment	Intersection (Left turn) risk assessment	Conclusion (Low risk < High risk)
Augmenting Reinforcement Learning With Transformer-Based Scene Representation Learning for Decision-Making of Autonomous Driving (H. Liu et al., 2024)	Collision rate: Min: 4% Mean: 39% Max: 100%	Collision rate: Min: 16% Mean: 40.56% Max: 100%	Collision rate: Min: 12% Mean: 32.66% Max: 54%	Merging < Roundabout Intersection (Left turn) < Roundabout Intersection (Left turn) < Merging
Deep Attention Driven Reinforcement Learning (DAD-RL) for Autonomous Decision-Making in Dynamic Environment (Chowdhury et al., 2024)	Collision rate: Min: 2% Mean: 31.25% Max: 100%	Collision rate: Min: 4% Mean: 34.17% Max: 100%	Collision rate: Min: 0% Mean: 25% Max: 54%	Merging < Roundabout Intersection (Left turn) < Roundabout Intersection (Left turn) < Merging

Table 4.4 contains two papers that use the same metric and include scenarios involving merging, roundabout, and intersection. (H. Liu et al., 2024)'s average collision rates are 39% for merging, 40.56% for roundabouts, and 32.66% for intersections. (Chowdhury et al., 2024)'s average collision rates are 31.25% for merging, 34.17% for roundabout, and 25% for intersection. Both papers show that roundabouts are the most dangerous, followed by merging, and lastly, intersections.

4.6. Comparative Analysis of Merging, Intersection and Highway

Table 4.5 Risk Assessment between Merging, Intersection and Highway

Paper title	Merging risk assessment	Intersection (Left turn) risk assessment	Highway risk assessment	Conclusion (Low risk < High risk)
Safety-constrained Reinforcement Learning with Interaction-aware for Decision-making of Autonomous Driving (Zhang et al., 2025)	Success rate: Min: 7% Mean: 44% Max: 94%	Success rate: Min: 12% Mean: 49.5% Max: 97%	Success rate: Min: 37% Mean: 60.75% Max: 100%	Intersection (Left turn) < Merging Highway < Intersection (Left turn) Highway < Merging

(Zhang et al., 2025) includes scenarios involving merging, intersections, and highways. The average success rate of the first paper was 44% for merging, 49.5% for intersection, and 60.75% for highway. This paper shows that merging is the most dangerous, followed by intersections, and lastly highways.

4.7. Analysis summary

Table 4.6 Summary of Scenario Risk Hierarchies from Primary Studies

No	Conclusion (low risk < high risk)	Support paper
1	<i>Intersection(Left turn) < Merging</i>	(L. Gao et al., 2025) (H. Liu et al., 2024) (Chowdhury et al., 2024) (Zhang et al., 2025)
2	<i>Roundabout < Intersection (Right turn)</i>	(Elallid et al., 2024)
3	<i>Roundabout < Intersection</i>	(Yun & Cao, 2024)
4	<i>Intersection (Left turn) < Roundabout</i>	(H. Liu et al., 2024) (Chowdhury et al., 2024)

Table 4.6 (continued)

5	<i>Highway < Intersection</i>	(Prasath A & Navaz K, 2025)
6	<i>Highway < Intersection (Left turn)</i>	(Hu et al., 2024) (Zhang et al., 2025)
7	<i>Merging < Roundabout</i>	(H. Liu et al., 2024) (Chowdhury et al., 2024)
8	<i>Highway < Merging</i>	(Zhang et al., 2025)

Table 4.6 summarizes the risk comparison findings extracted from all studies, which forms the basic logical elements for subsequently constructing an overall scenario hierarchy. Each entry follows a formal relational logic, with the left side representing perceived risk or complexity lower than the right side. As shown in the table, empirical evidence from recent literature.

Table 4.7 Logic Synthesis of Driving Scenario Complexity Hierarchies

Logic Components	Synthesized Hierarchy
$(I(l) < Rab) + (I(l) < Mrg) + (Mrg < Rab)$	$I(l) < Mrg < Rab$
$(H < I) + (H < I(l)) + (H < Mrg) + (I(l) < Mrg)$	$H < ((I(l))/I) < Mrg$
$(H < (I(l)/I)) < Mrg) + (I(l)) < Mrg < Rab)$	$H < (I(l)/I) < Mrg < Rab$
$(H < (I(l)/I) < Mrg < Rab) + (Rab < I(r)) + Rab < I$	$H < (I(l)/I) < Mrg < Rab < (I(r)/I)$

This chapter systematically constructs a hierarchical model of risk and complexity in autonomous driving scenarios through qualitative analysis of eight primary studies.

Based on the literature evidence listed in Table 4.6, this study extracts direct comparative data on relative risk levels under different traffic environments. The study finds that although different scholars have different research focuses, they show a high degree of consistency in core conclusions such as "highways

have the lowest complexity" and "intersection turning tasks have high complexity."

Through the derivation process of the logic components shown in Table 4.7, this study uses transitive inference to synthesize fragmented evidence into a complete hierarchical architecture. The final synthesized hierarchical model (Highway < Intersection (Left turn) / Intersection < Merging < Roundabout < Intersection (Right turn) / Intersection) clearly characterizes the gradient of task challenges faced by autonomous driving systems in the operating domain.

The scenario hierarchy model established in this chapter not only verifies the consistency of existing literature, but also provides theoretical support and scenario selection basis for the research on decision-making algorithms for high-complexity scenarios (such as right-turn intersections) in subsequent chapters.

Chapter 5: Transformer Architectures in Autonomous Driving

5.1. Introduction

In autonomous driving, there are various variants based on the Transformer, each designed to address specific shortcomings, such as training efficiency or suitability for different perception modalities. The following are some of the more common Transformer variants used in autonomous driving.

5.2. Vision Transformer

Table 5.1: Papers that use the Vision Transformer model

Paper title
A Preference-Based Multi-Agent Federated Reinforcement Learning Algorithm Framework for Trustworthy Interactive Urban Autonomous Driving (Lu et al., 2025)
Deep Reinforcement Learning for a Self-Driving Vehicle Operating Solely on Visual Information (Lu et al., 2025)
Efficient Maximum Entropy Reinforcement Learning Framework Guided by Expert Policies: An Application in Autonomous Driving (L. Gao et al., 2025)
Improving Reinforcement Learning with Expert Demonstrations and Vision Transformers for Autonomous Vehicle Control (Elallid et al., 2024)
Integrating Driving-Aware World Model With MPC for Autonomous Driving at Unsignalized T-Intersections (X. Zhang et al., 2025)
Vision Transformer for Learning Driving Policies in Complex and Dynamic Environments (Kargar & Kyrki, 2022)

Vision Transformer (ViT) is a model that directly applies the Transformer architecture to image classification tasks. It segments images into fixed-size patches, linearly embeds these patches into a one-dimensional sequence, and then feeds it into a standard Transformer encoder for processing. ViT performs well in multiple computer vision tasks and is widely used in the field of autonomous driving for end-to-end perception and decision-making tasks.

(Audinys et al., 2025) installed a camera in front of a virtual vehicle to capture 224×224 pixels RGB images. In order to enable the model to infer the vehicle's current motion state (such as speed and acceleration), the input was not a single image, but rather four consecutive frames stacked together as a set of inputs.

(G. Li et al., 2025) used the Swin Transformer, a model with a hierarchical structure and a shifted window mechanism compared to ViT. This design allows the model to recognize both macroscopic objects and subtle local details in images, similar to a CNN, making it well-suited for object detection and semantic segmentation tasks. The image is divided into multiple non-overlapping local windows (e.g., 4×4 or 7×7 patches), and self-attention computation is performed only within each window.

5.3. Decision Transformer

Table 5.2: Papers that use the Decision Transformer model

Paper title
Autonomous Intersection Management via Prior-Enhanced Multi-Agent Constrained Decision Transformer (R. Zhao et al., 2025)
Deep reinforcement learning navigation via decision transformer in autonomous driving (Ge et al., 2024)

Decision Transformer (DT) is a model that uses a transformer architecture to decision-making in reinforcement learning (RL). Contrary to policy gradient-based reinforcement learning, DT views the sequential decision-making problem as an autoregressive sequence modeling problem. It is a direct predictor of the future action sequence based on a sequence of states, actions and outcomes hence goal-directed decision-making.

According to (Shadab Siddiqui et al., 2025), DRLNDT has better perceptual accuracy with a success rate of up to 99.9 percent in navigation exercises. Moreover, the decision transformer architecture is able to process sequential positional data in parallel, which is much faster and more effective in training in complex environments.

5.4. General Transformer

Table 5.3: Papers that use the General Transformer model

Paper title
Attention-Based Distributional Reinforcement Learning for Safe and Efficient Autonomous Driving (J. Liu et al., 2024)
Balancing Individual and Collective Interests: Integrating Transformer-Based PPO and Social Intelligence for Autonomous Vehicles at Unsignalized Intersections (Doroudian & Taghavifar, 2024)
Empirical Evaluation of Deep Learning based Autonomous Driving System with Real-Time Scene Classification Procedure (Prasath A & Navaz K, 2025)
HGRL: Human-Driving-Data Guided Reinforcement Learning for Autonomous Driving (Zhuang et al., 2024)
Improving policy training for autonomous driving through randomized ensembled double Q-learning with Transformer encoder feature evaluation (Fan et al., 2024)
Multi-Modal Attention Perception for Intelligent Vehicle Navigation Using Deep Reinforcement Learning (Z. Li et al., 2025)
Pre-trained Transformer-Enabled Strategies with Human-Guided Fine-Tuning for End-to-end Navigation of Autonomous Vehicles (Hu et al., 2024)
Safety-constrained Reinforcement Learning with Interaction-aware for Decision-making of Autonomous Driving (Zhang et al., 2025)
SoftCTRL: Soft conservative KL-control of Transformer Reinforcement Learning for Autonomous Driving (Huynh & Nguyen, 2024)
Stability-Aware Reinforcement Learning for Autonomous Driving With Dynamics-Augmented State and Lyapunov Constraints (Luo et al., 2025)
TSAC: Transformer-based Soft Actor-Critic for Behavior Decision-making in Autonomous Driving (Z. Gao et al., 2025)
Autonomous Driving Agents for Safe Road Junctions Through Deep Reinforcement Learning (Yun & Cao, 2024)

The general Transformer architecture is used as the core component of the state encoder or policy network in most studies. These are the models that are built on the typical Encoder-Decoder or encoder-only architecture but tailored to the autonomous driving tasks, e.g., by adding multimodal fusion mechanisms, e.g., CNNs and LSTMs.

These models do not only receive one form of data input, like only visual

data, but instead combine data of a variety of sensors, like vehicle posture and velocity, or lidar data.

(Prasath A & Navaz K, 2025) is a combination of Transformer and CNN models. It accepts three kinds of data as input: Ego State (steering angle, velocity), Navigation Info and Surrounding Info (lidar). Since the three inputs are of varying attributes, the system applies three independent fully connected networks (FCNs) to process the ego state, navigation, and surrounding information respectively, and convert them into feature vectors of the same length (e.g., $L=256$).

5.5. Other Transformers

Table 5.4: Papers that use the Other Transformer model

Paper title	Transformer model
Augmenting Reinforcement Learning With Transformer-Based Scene Representation Learning for Decision-Making of Autonomous Driving (H. Liu et al., 2024)	MST + SLT Transformer
Constraint-Guided Behavior Transformer for Centralized Coordination of Connected and Automated Vehicles at Intersections (Zhao et al., 2024)	Behavior Transformer
Deep Attention Driven Reinforcement Learning (DAD-RL) for Autonomous Decision-Making in Dynamic Environment (Chowdhury et al., 2024)	Spatio-Temporal Attention Encoder
HGRL: Human-Driving-Data Guided Reinforcement Learning for Autonomous Driving (Zhuang et al., 2024)	GPT2
MTD-GPT: A Multi-Task Decision-Making GPT Model for Autonomous Driving at Unsignalized Intersections (Liu et al., 2023)	GPT2

Transformer architectures have also been suggested to be customized or optimized to particular autonomous driving tasks, or other pre-trained language models (including GPT-2) can be used as decision encoders.

Both (Zhuang et al., 2024) and (Liu et al., 2023) used a generative pre-training architecture that is based on GPT-2 to process decision-making in

autonomous driving. MTD-GPT centers on the enhancement of the accuracy of decision-making in intersections without traffic lights using multi-task learning, whereas HGRL additionally uses human expert information and uses the causal modeling power of GPT-2 to maximize the stability and efficiency of reinforcement learning.

To improve the scene representation and accelerate the training process, (H. Liu et al., 2024) applied MST and SLT Transformers. The Behavior Transformer was applied to solve the problem of multi-vehicle conflicts and absence of rule constraints (Zhao et al., 2024). Spatio-Temporal Attention Encoder was used in (Chowdhury et al., 2024) to overcome high computational complexity and spatiotemporal interaction.

Chapter 6: Implementation Plan

6.1. Development Environment

The training process will utilize significant GPU resources. I will use GPU resources provided by the MMU and the local server's GPU for training. The MMU's GPU will primarily be used for training and testing, while the local GPU will be used for the CARLA simulator. Both will be connected using the Cloudflared TCP tunnel protocol.

6.2. System Architecture Design

The entire training system will be divided into three modules: a simulator, a perception module, and a decision algorithm. The simulator will provide sensor data and execute operational commands such as throttle input. The perception module will use a Transformer encoder to process this multimodal data, such as data from RGB camera sensors and LiDAR. The decision algorithm implements a soft Actor-Critic algorithm for risk perception.

6.3. Scenario Selection

This research focuses on the critical high-risk scenario of intersections and simulates it in the CARLA environment. To accelerate the agent's learning, I designed a training process with increasing difficulty. The agent first trains in low-difficulty scenarios, learning basic driving skills. Medium difficulty increases traffic density, pedestrian numbers, and inclement weather. High difficulty introduces unexpected events.

6.4. Evaluation Metrics

Performance evaluation employs a comprehensive multi-dimensional metric framework. Primary safety metrics include collision rate. All evaluations follow standardized protocols with statistical significance reporting and both quantitative metrics and qualitative assessments.

Chapter 7: Conclusion and Outlook

7.1. Summary of Research

This study systematically reviews academic literature from 2019 to 2025, exploring the safety performance of Deep Reinforcement Learning (DRL) algorithms based on the Transformer architecture in high-risk autonomous driving scenarios.

- **Algorithm Performance:** The study found that although there are some differences in the results among different studies, in most comparative experiments, SAC (Soft Actor-Critic) showed a lower collision rate and a higher task success rate than PPO and DDPG. However, the performance of an algorithm largely depends on the specific task attributes and environmental randomness; there is no absolutely "best algorithm."
- **Scenario Risk Level:** Through logical derivation and comprehensive analysis, this study established a scenario complexity hierarchy model. The results show: Highway < (Intersection (Left turn) / Intersection) < Merging < Roundabouts < (Intersection (Right turn) / Intersection). This demonstrates that turning at unsignalized intersections remains the most severe safety challenge currently facing autonomous driving.
- This literature review found that most publications tend to use the General Transformer as the core state encoder or policy network. These models, by introducing a multimodal fusion mechanism, can effectively integrate heterogeneous data such as vehicle attitude, velocity, and LiDAR.

7.2. Future Outlook

Based on previous literature review, the next phase of this Final Year Project 2 (FYP2) will move towards practical simulation experiments:

- **Experimental Platform:** The simulator will be based on CARLA simulator, utilizing GPU resources provided by the MMU for intensive reinforcement learning training. An alternative is Google Colab.
- **Key Scenarios:** The selection of algorithms, models, and scenarios for FYP2 will be based on the findings of this literature. The algorithm will use SAC, the model will be Transformer, and the scenario will be Intersection.
- **Core Objective:** FYP2 will focus on developing and training an agent that integrates Transformer and SAC algorithms. The goal is to minimize the collision rate of vehicles in Intersection scenarios to improve safety.